1 (a) Write 72 as a product of its prime factors.

$72 = 2 \times 36$

$= 2 \times 2 \times 18$

$= 2 \times 2 \times 2 \times 9$

$= 2 \times 2 \times 2 \times 3 \times 3$

$72 = 2^3 \times 3^2$

1 mark
Give a product of primes.

1 mark
Keep dividing
until all factors are prime.

$2^3 \times 3^2$

(2)

(b) Find the highest common factor (HCF) of 72 and 108

$72 = 2^3 \times 3^2$

$108 = 2^2 \times 3^3$

Common prime factors: $2 \times 2 \times 3 \times 3$

$HCF = 2^2 \times 3^2 = 36$

1 mark
Take the smaller power of each shared prime.

1 mark
Decompose 108 too.

36

(2)

(Total for Question 1 is 4 marks)

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